Or the degrees of freedom, talked about some important properties of the kai square distribution. It has mean equal to m, the degrees of freedom, variance equal to 2M. It has this nice addativity property. In particular, this second piece, the converse, if x3 is kye squared, its degrees of freedom is m1 plus m2, where x1 and x2 are both kai squared with corresponding degrees of freedom, m1 and m2, and x3 is the sum of those two, x1 plus x2. then x1 and x2 must be independent. So we're going to use this property today. We saw it's the, it can be generated as the sum of independent squared, standard normal random variables, right, Z1 squared plus Z2 squared, so on. Did some visualization here, just looking at the probability density function for different values of M, that degrees of freedom parameter. We see it's very, very skewed when M is small, right? But it approaches a normal distribution as M gets large. That's the result of the central limit theory. We saw that's related to the gamma distribution, specifically it's a special case of the gamma distribution, where that alpha parameter is m over 2, and that beta parameter is just fixed at 1/2. And we saw if we take a kai squared random variable x, and we scale it by some constant c. And the result is also a gamma random variable, and c comes into this beta parameter of the gamma distribution. X follows a standard normal distribution, the random variable x squared, Follow the kai squared distribution with one degree of freedom. And then we just ended with a couple examples. One thing we showed is when we have a random sample from a normal distribution where the mean is known, but the variance is unknown. Then the MLE for the variance is this. And we showed the scaled version of that MLE, so scaling it by n over sigma squared, but this will follow a kai squared distribution n degrees of freedom. Now again, this was for the case where mu was known. We're going to talk today about when mew is unknown, but this will come back up again. We'll use this fact. It'll help us today. And then just a couple other examples there. Okay, so today we're going to talk about, again, this case where we have a sample from a normal population. Normal distribution comes up all the time, right? A lot of things naturally follow a normal distribution, or at least are well approximated by a normal distribution, things like heights and test scores and yields of crops and so on. So it comes up a lot. So we want to understand the sampling distribution of our estimators in the case where we have a normal population. And in particular, we typically will not have, say, a known mean or a known variance. We'll typically want to estimate both of those parameters. And so that's what we're going to talk about today. Talk about kind of the joint sampling distribution of the sample mean and the sample variance. That's the goal of today's lecture. And so again, we're in this setting where we have a random sample from a normal distribution with these parameters, mu and sigma squared, only now both of these parameters are unknown. In this case, our maximum likelihood estimators for these two parameters are the sample mean mu hat or x bar. That's our maximum likelihood estimator for the mean mu, and then this maximum likelihood estimator for the variance, right, the average of these squared deviations now around our sample mean, our mu hat. Since we don't know the true mean mu, right, we have to estimate it by plugging in our sample mean mu hat.

So these are both random, right? They're a function of the random data x, so they will be random. It will vary from one possible sample to the next. So they have some distribution, and we want to understand that distribution, the sampling distribution. And since we're talking about more than one quantity here, there's two things. This is a joint distribution, specifically a bi-variate distribution. There's two random variables here. There's mu-hat and sigma-square-hound.

We want to understand how these statistics, these estimators behave. I'm going to understand their joint distribution when we sample from a normal population and we have these as our MLE of those two parameters, mu and sigma hat. This is the important piece today, this theorem here. So we have this setting. We have data X1 through XN. Their IID are a random sample from a normal distribution, a normal population, parameters mu and sigma squared.

So in that setting, we have. have these three facts. The first fact is that mu hat, our MLE for the mean, which is just calculated as X bar.

Its sampling distribution will be normal.

Distribution of X bar is normal. Its mean is mu, so it's centered around the true parameter mu, and its variance is sigma squared over n, so it gets smaller as N gets larger. It's the first fact.

The second fact involves the MLE of the variance, sigma squared hat.

Remember we just showed was this average of these squared deviation, around the sample mean x bar. And the scaled version of this, just like on our last lecture, we were scaling sigma squared hat by n over sigma squared. We're going to do the same thing here. Only sigma hat squared is different because we can't plug in the true value here. We have to plug in x bar our estimate from mu. This is a little bit different, but we're still scaling it the same way. And it still follows a k-squared distribution. Only now we have n-minus 1 degrees of freedom rather than n. So that's the second fact. We plug in x-boread. bar instead of mu, since we don't know mu now, and the resulting scale distribution of this is kai squared with n minus 1 degrees of freedom. And then the last fact is that these two things are independent. X bar are mLE for mu, and sigma squared hat are mLE for the variance, are independent random variables.

Yeah, question. So the mle for variance by itself doesn't follow kai square distribution, really the scaled version of it does?

Correct, yeah. The scaled version follows a kye square distribution, and we know if we scale it by multiplying by its inverse, it'll be gamma. So that's basically it. This is the point of today's lecture. If you take nothing else away, just remember these three things. This is what I want you to learn today. So I'm going to spend the rest of the lecture just trying to convince you that these three things are true. I will do that partially by proving these, partially by doing some simulations, simulating data, and calculating these random variables, X bar. and Sima Squared Hat and just trying to show you visually that these properties hold. But again, this theorem here is the point of today's lecture. Any questions on this? Along the way, we're going to use this other theorem just to help us. This is just really kind of a helper theorem to prove one of the results we'll need. It's not too interesting on its own, but here this says, suppose Z1 through ZN are IID, standard normal now, standard normal random variables. So we know the mean is zero, and we know the variance is one. There's no unknown parameters here. We have standard normal random sample, then this quantity, the sum of these squared deviations around the sample mean, Z bar, follows a kai squared distribution and minus one degrees of freedom. I'll go ahead and sort of show this real quick, but again, we're just going to use this for something later on. It's not really something we're interested in on its own.

So we're going to have a little fact that's going to help us here.

Full fact. If x is a random variable, kye squared random variable, n degrees of freedom, y is another random variable, and it's kai squared with one degree of freedom, x is equal to y is w, and one more condition, y independent w.

If you aren't familiar with this symbol, it just means independent. Okay, so y is independent of w. If we have all of these conditions, then this random variable w has to be kye squared with n minus one degrees of freedom. Just something I'm giving you, and we're going to use it in a moment. But we're interested in the sum here, the sum of these squared deviations or residuals around the sample mean. Okay, to write this as sum, I equals 1, n, zi, bar, squared. And we're both basically going to expand each of these squares. So for each one, we're going to have a zi squared, sum, zi squared. We have 2ab, or minus 2 zi, z bar, and we have a sum, so minus 2, zi, z bar. z bar summation z i the cross term and then for each one we'll have a z squared term plus n z bar squared just expanding those squares if you look at this term in parentheses the sum of z i this is equal to n times z bar taking the definition of z bar and

multiplying by n so that's equal to n z bar so this is minus two z bar squared or minus two n z bar squared and we have plus one n z bar squared together with these two terms, we have minus n z bar squared.

So this equation simplifies to sum z i squared and s z bar squared. So we have this red term, the sum of the squared deviations is equal to this green term, the sum of the squared z's minus this purple term n z bar squared.

So let's look at this green term for a second.

Since z's iID standard normal, and that was the setup we had here, They're IID standard normal. What do we know about ZI squared? We take one of these Zs and we square it, what distribution will that have? Chi squared with what degrees of freedom? One. Yep. So each of these ZI squared will follow kai squared distribution with one degree of freedom. And if we add them up, sum, zI squared, what is that? If we go to lecture 8 and in our properties here, one of our properties is that it's the sum of independent squared standard normal variables. So if we have M independent standard normal variables, we square them, we add them up, that's kai squared with m degrees of freedom. So if we just have one that we square and add up, it's going to have one degree of freedom.

Each one is going to be one degree of freedom, but if we add up more than one, we're sort of accumulating degrees of freedom.

Okay, so here if we have n of these, we square them, we add them up. What distribution is that? Chi squared with degrees of freedom.

Know the distribution now of this green term.

Yeah. question, but when we're like expanding out like the square, so you did a, you know, a squared minus 2ab plus 2 a b. Yeah. So how come for the z bar you did times two, but not like two times n? Because for the z bar, when, for like the b squared term, you did n like the z bar squared.

So here we have z bar squared, and we have one of those for every i. There's n of them.

So we have n z bar squared. Here, right, We're just pulling the 2 in the Z bar outside of the summation. And then the summation, what's left, sum of ZI is NZ bar. So it reduces to 2N Z bar. Okay, so we know the distribution of this green term now. It's kye squared with n degrees of freedom. Okay, let's look at the purple term now. Z bar, be normal, mean zero, variance 1 over n. Okay, so let's transform this, multiply it by square root of n, square root of n times z bar. That's the distribution of this going to be. It's going to be normal. It'll still have mean 0 because the expectation of z bar is 0. And what will the variance be? So we have constant times a random variable. What's the variance of a constant times a random variable? It's 1. So if we have a constant inside a variance, we square it and pull it outside of the variance. So we have an n. And then the variance is 1 over n. So we have n has 1 over n is 1. So n z bar will also be normally distributed, mean 0, variance 1, or standard normal. Then if we square it, square root of n, z bar, whole quantity squared, which is just n z bar squared, what distribution will this have? Chi squared with how many degrees of freedom? We know the distribution. The green term, it's kai squared n. We know the distribution of the purple term. It's kai squared one. And know what the red term is equal to the green term minus the purple term, or just rearranging that, purple term, n bar squared, plus the red term, summation, squared. equal to our green term, the summation of the squared standard normals. This, I squared with one degrees of freedom, know this pi squared n degrees of freedom, really said what this red term is yet. We know purple plus red is green. We know the distribution of purple. We know the distribution of green. This is where we can start to use this fact. So take the purple term, consider that Y in this helpful fact. Consider the red term to be W. Consider the green term to be X. W. Okay, we know X is equal to Y plus double. established, you

know y is kye squared 1, it's established, we know x is kye squared n, that's established. The only thing we haven't said yet is that y is independent of w. If we have that, then we'll know that w is kai squared minus 1 degrees of 3. Now we know z bar is normally distributed. We know these deviations zi minus z bar are normally distributed. We have a normal random variable. Does anybody remember how we can check? two normal random variables how we can check if they're independent. I mean by the intersection. Special property of the normal distribution has to be normal. No, we're trying to check if two normal random variables are independent. Stat 131. If they are independent, then the variances do some. How we can check for independence is to look at the covariance. So specifically for the normal distribution, if the covariance is zero, then they're independent. So that's what we can check. Let's look at covari, z bar, zi, minus, Z bar.

Properties of covariances, we can do this as covariance of these two terms, covariance of those two terms, covariance, z bar, the I, and the second term will just be minus the variance of z bar.

For this first term, z bar is 1 over n times the sum of all our z's.

And we know, because our z's are iID, any given z is independent of all the others.

So the only one in this z bar that's going to be dependent on zi is the zi term.

We're doing. to the covariance of 1 over n z i 1 over n the only z term in the sum will be dependent minus the variance of z bar it's this covariance just 1 over n yeah so for variances we square the constant when we pull it out for covariances you just pull it out so it's 1 over n so you can imagine pulling the 1 over n out and you have covariance of zi and zi which is just the variance of zi or 1 the first term is 1 over n the variance of z bar we have 1 over n minus one over n is zero.

So Z-bar is independent of any of these residuals, these deviations around the sample mean.

Since z-bar, z-bar, z-I-minus z-bar, are both normal.

Zero covariance lies independent. The first one is the covariance equation you set up. Is that like a formula, like a standard formula where the covariance of two variables equals the covariance minus the variance? No, it just comes from covariance of x and y plus z to the covariance of x and y, the covariance of x and y, the covariance of x.

It's just that covariance decomposition. Now, in the second term, you would have minus covariance of z bar and z bar, which is the variance of z bar. So the covariance of anything in itself is the variance of that thing. How we know that z bar has a variance of one over n? That's something we proved earlier? Sort of.

We showed it in our sampling distribution lecture. We showed it for kind of arbitrary mu and sigma squared. We said it was sigma squared over n. In this case, it's standard normal, so sigma squared is one, so one over n. But yeah, how we got there was just the variance of mu hat or x bar is the variance of this sum of each random variable over n.

Because of independence, we can write that as the sum of variances, and the one over n terms square when they come out of the variance, everything up to here would be the same, and then in our case, variance of each x would be one, so we'd have one over n.

Okay, so zi, and any of these deviations are independent. Purple term, bar squared, is just a function of z bar, must to be independent of some of our squared deviations, which are just a function of the deviation. So again, remember from stat 131, if two random variables are independent, then any

function of those two random variables are also independent here. Our purple term and our red term are independent.

And since they're independent, we get from our helpful fact that this red term, some of our squared deviations around the sample mean is kai squared n minus one degrees of freedom.

Notice here, some of the individual standard normal random variables squared is kai squared n.

When we first center them, subtract z bar, we lose that degree of freedom, right? We go to n minus one instead of n. We're losing a degree of freedom. And if you read the reading for this week, you saw that that's basically because of the fact that the sample mean, z bar here, if we know the sample mean z bar and we know the first n minus 1 z, there's no freedom for the last z to be anything other than the value that makes it the sum. We're losing a degree of freedom. There's no flexibility for that last random variable to vary given that we know z bar. So that's the idea there. We're losing a degree of freedom, estimating or centering on this sample mean. Z bar here.

So again, this is just a fact that's going to help us in a bit. That was the first thing I wanted to show this one.

Okay. So that's going to help us, but the point of today's lecture is to go through these three properties from our first theory. I start with this first one, the distribution of the sample mean, X bar.

Again, we're in the setting where we have X1 through XN, a random sample IID, a normal population, mean mu, variance in a squared. The first thing we want to show is the x bar also follows a normal distribution. Its mean is mu, and its variance is sigma squared over n. And we actually showed this in the sampling distribution lecture, the thing I just pulled up a minute ago, but I will go ahead and just go through that again. Okay, so we're talking about the sample mean here, x bar, x bar, which is just x1 for n, two over n, and so on.

The sum of our random variables x divided by the samples of n. This is a weighted sum of normal random variables, sum, normal.

If we have a weighted sum of normal random variables, what family of distributions, must that result be in?

Squared normal is just normal. Anytime we take a linear combination or a weighted sum of normal random variables, the result will also be normal. X bar must be normal.

I've shown this part already. It's normal. We just need to show now that it has mean mu and variance, same with squared over in, the two parameters of the normal distribution. Well, the mean, mu, is just the expectation.

Go ahead and find the expected value of x bar, the mean, and that's just the expected value of this sum, x1 over n, 2, or n, and so on.

The definition of x bar being plugged in there. And one of our properties of expectation is linearity.

Linearity of expectation says the expected value of a sum is the sum of the expected value. So we can write this as expectation of x1 over n.

So just pulling the constant outside of each expectation.

It's the expectation x2. over n and so on through the last term expectation of xn over n each of these expectations is what what's the the value of the expectations me yeah each of these x's has expectation mu yep so we have mu over n and we have n of those terms in terms that are each mu over n gives us mu so

we know x bar is normally distributed with mean mu now and then the last thing we need is the variance the variance of x bar again just plug in our definition for x bar here.

It's one for n, two over n, and so on. We know these are independent. What does independents let us do with variances?

Yeah, they're linear. We're not have to worry about the covariances essentially.

The covariances are all zero. So the variance of a sum will also be the sum of the variances in the special case where we have independent. I could write this the variance, the variance, two over n, and so on.

Okay, each of these terms has a constant one over n. When we pull a constant outside of the variance, we square it. of these terms squared, and what is the variance of each of these random variables at sigma squared?

They're all normal, mean, mu, variance sigma squared, so they all have variance sigma squared, n terms that are all sigma squared over n squared, which reduces to sigma squared over n.

Okay, so we know it's normal, we know its mean is mu, we know its variance is sigma squared over n, but we know the full distribution of x bar, x, x, bar, normally distributed mean mu, variance sigma squared over n. So you're saying we did this exact same proof in lecture 7? Yep, yeah. We did this same thing in lecture 7. Now we're just rehashing it because it's part of the joint distribution for the sample mean and variance. So that's proving fact 1 to you, and now I'll show it through simulation. So here I'm taking repeated samples. Each sample is going to be size 10. And is equal to 10. And I'm taking a sample size 10 from a population that's normally distributed with mean 50 variance 4. So mu is 50, sigma squared is 4. And each time I take a sample of size 10, and I'm going to compute the sample mean. I'm going to compute x bar. I'm doing this many, many times. I take a sample of size 10 from this population. I compute the sample mean, I throw that sample mean in a bucket. I take another sample size of size 10. Compete the sample mean, throw that in a bucket. And I'm going to look at the distribution, empirically, of this bucket of sample means that I've got. That's the sampling distribution of the sample mean. And we just said that sample mean should have this sampling distribution. It should also be normally distributed. It should be centered around true mean mu, which is 50 in our case. And it should have variance sigma squared over n, which is 4 over 10 in our specific. It should have that sampling distribution. If I repeat this many, many, many times and look at the distribution of these sample means, it should look very close to that specific normal distribution. That's what I do here. In blue is the histogram of these sample means. And then this red curve here is the probability density function for this normal distribution with mean 50 and variance 4 over 10. Theoretical sampling distribution. And we see they match up very, very nice.

If our sample size had been bigger, if we had said n is equal to 100 instead of 10, right, this would still be normally distributed. It would still be centered around 50, but the variance would then be 4 over 100, much smaller. So we'd look at something much more concentrated than what we have here.

This is the theoretical distribution and the histogram matches up very nicely here.

Yes. So if you took infinite samples, infinite times of this sample 10, compute x bar, throw it in the bucket. If you do that infinite times, the long run average would be 50. Now for any finite number of these that you do, it's not going to be exactly 50, but it's going to be very, very close.

10,000. So any questions on the first property? The next property, property two of our important theorem here, concerning the sample variance.

Again, we're in this situation. We have x1 through xN. It's a random sample from a normal population with mean, me, meo, and variance.

And we're looking at the distribution of this scaled MLE for the variance. The MLE for the variance is sigma squared hat, but we're scaling it. We're multiplying it by n over sigma squared. And we said that scaled version follows a kai square distribution with n minus 1 degrees of freedom.

So we know sigma squared hat, the sum of our squared deviations on the sample mean. But we don't want n over sigma squared hat. We want n times sigma squared hat. We want n times sigma squared hat divided by squared show or sigma squared some of these squared deviations around the sample mean with this as well as a kai squared distribution with n minus one degrees of freedom. So remember, whenever we have something that's normally distributed, we can take a z score, subtract the mean, divide by the standard deviation. And we can then get something that's standard norm. z i muted by taking each x i minus mu over sigma this will be standard normal and we're going to want to work with standard normal so that we can use this theorem earlier here we were working with standard normal where we we know the mean zero we know the variance is one but that's not the case in actuality so we're going to take z scores okay so z i is x i minus me over sigma that's standard normal and we're just going to rearrange this equation solve for x essentially here x i is mu sigma z i and you can do the same for x bar and z bar x bar is going to be mu plus sigma z bar again z bar it's just one over n the sum of the z okay so we've got x in terms of z x bar in terms of z bar so then our sum of squared deviations sum x i minus x bar squared we're just going to take this replace x i with our representation of it into z replace x bar with mu plus sigma z That's all we're going to do. The summation xI or mu plus sigma z i, this x bar or mu plus sigma z bar whole quantity squared squared squared deviation. Notice our mu's cancel. This is just going to reduce to the summation sigma z i minus sigma z bar squared, which is equal to summation sigma squared and z i minus z bar squared. You have some of squared deviations around the sample mean to these sum of squared deviations of z around their sample mean times sigma squared.

We're just going to divide both sides by sigma squared now.

So then we have one over sigma square. We're just dividing both sides by sigma squared. One over sigma squared.

Our summation squared deviations of x around its sample mean is just equal to sum of z i minus z bar squared.

One over sigma squared times the sum of our squared deviations of x. That's what we want. We've shown it's equal to this. What's the distribution of zi minus z bar squared sum? Chi squared and minus one.

That was that helper theorem that I gave you earlier, this right here. That's why we showed that it was just a step along the way to this proof. So we showed that the thing we're interested in is equal to this and this thing is kye squared with n minus one degrees of freedom. This kai squared and minus one freedom distribution 2 above.

from our theorem 1. Questions on this proof? Yeah. that n-sigma-squared hat over sigma-squared follows a k-square-n distribution. We're just trying to show then that a scaled version of sigma-hat squared just follows kyn-square. So we had different definitions of sigma-squared hat there. So in the lecture 8-1, notice we were plugging in the true mean mu. Here, we don't know the true mean mu, so we are plugging in mu-hat or x-bar. We're estimating mu. Part of that process of estimating it causes us to lose a degree of freedom. Okay, so this second property, the distribution of this scaled MLE for the variance, shown that mathematically. Again, I'm going to show this through a simulation. Here we're going to take repeated samples, this time of size 15. We'll do a different normal population. This one will be centered at 20, mean 20, variance 9. And for each of these samples, we're going to compute that MLE of the variance, or the

scaled version of it anyway. N equal to 15 over sigma squared 9 times our MLE, sigma squared hat, or sigma squared hat plugging in our estimate for me. It's plugging in x bar. We don't actually know x bar in that. We just said that this should have kai square distribution with n minus 1 degrees of freedom, or 14 degrees of freedom. That's the theoretical sampling distribution here. So again, we'll take a bunch of samples. We'll compute this statistic or this T here, and we'll compare the empirical distribution of those to the theoretical distribution, this kai squared 14 probability density function. Again, we see they match up very nicely. Again, I think I took 10,000 samples, yep. And the more and more you take, the closer and closer, this will get to the true probability density function. Any questions on our second fact from our theorem? So we know the distribution of x bar, the MLE for the mean, is normal, mean mu, variance sigma squared over n. We know that the scaled MLE for the variance follows a kai square distribution with n minus 1 degrees of freedom. And then the last fact is independence. We said X bar and the MLE for the variance are independent. If you read the textbook, it does show this as well. It does it in a bit of a different way. So it uses orthogonality and linear algebra to show this. That's not a prerequisite for this course. That's not how I'm going to do it. But if you're interested in that at all, please read the section on the textbook. to see an alternative way to do this proof. Again, we're in this setup where we have x1 through xN, IID, random sample from a normal population, mean mu, variance sigma squared, and we want to show that the MLE for the mean, x bar, and the MLE for the variance, similar squared hat, are independent. That's our goal here. So remember from our last lecture, our last lecture, we had some of squared deviations around the true mean, sum of xI minus mu squared, and what was the distribution of this? chi squared with what degrees of freedom, n degrees of freedom. So if we knew the true mean mu here in these deviations, it would be kai squared. We just showed if we plug in x bar as our estimate from you, we lose a degree of freedom, it's n minus 1. But in this case, where we know mu, it's n degrees of freedom. Let's write this sum squared deviations, it's mu squared. So there's a common trick that you'll see if you take later statistics courses. You basically take the thing you're missing, x bar, and you add it and subtract it. So we can write this as summation xI minus x bar plus x bar minus mu squared. So we didn't change anything, add and subtracted the same value. They cancel out, it's like adding zero. But now we can view this as two terms. Term a in this square is x i minus x bar, and our term b in this square is x bar minus mu. Okay, so now when we expand this square, we do a squared plus two a b, is b squared, summation, our first term, A squared, xI, x bar squared, to ab to summation, to summation xI, x bar, minus mu, plus our b squared term, x bar minus mu squared. Notice that this term doesn't involve i, we could pull it in front of the summation, and then we'd be left with the summation of xI minus x bar. Anybody want to guess what the summation of xI minus x bar is? What is that? Zero. And so we have the summation of xI minus x bar, which is equal to. to summation of x i minus n x bar we distribute the sum we have summation of x i and then x bar doesn't vary across i so we have n of these x bar terms the summation of x i is n x bar just rearranging the definition for x bar so this is equal to zero so this middle term goes away when we expand our square and we're left with summation of x i x bar squared again these terms don't involve i so they're all going to be the same we have n of them so plus n x bar minus mu squared.

So we have our sum of squared deviations around the true mean is equal to the sum of the squared deviations around the sample mean, I think we actually care about, plus this other term involving the deviation of our sample mean around the true mean. Left-hand side, right-hand side, we're actually interested in this over sigma-squared, so we're going to divide both sides by sigma-squared. Sum of x-I yes mu squared or sigma-squared, the right-hand side over sigma-squared, squared, squared deviations, the sample mean over sigma squared, n x bar is mu squared over sigma squared. So again, we have the red term is equal to the blue term plus the green term. Let's start looking at these terms individually. We already found the distribution of this one. What was it? Chi squared with n minus one degrees of freedom. That was our second property from our theorem. Earlier, summation of our squared deviations in the sample mean over sigma squared, kai squared distribution, n minus one degrees of freedom. We

already have that. Look at the green piece now. Start with what we know. We see x bars in there. We know the distribution of x bar. That was our first part of our theorem. So we know x bar normally distributed. Its mean is mu. Its variance is sigma squared over n. So then we take x bar and we subtract its mean, mu, and we divide by its standard deviation. What distribution will that follow? Standard normal. X bar minus mu over its standard deviation. the square root of sigma squared over n, this will be standard normal, or normal 0.1. What if we square that?

What distribution would it follow? Chi squared with what degrees of freedom? One.

Yeah, if we square a single standard normal random variable, it's one degree of freedom. Square root of n, it's bar, this mu, this mu, sima. If we square this quantity now, simplify n x bar, this mu squared, or sigma squared. We square this quantity, it will follow a kai squared distribution with one degree of freedom. Okay, so the blue term is kye squared n minus 1, the green term, I squared 1, let's call the red term x3. So x3, sum, squared deviations around the true mean mu squared, the blue term x1, psi squared, squared, and I'm going to call this green term, x2, x bar squared, or sigma squared. And what do we know? Well, we know the red term, x3, all the kai squared distribution with n degrees of freedom, that as n minus 1 plus 1. It's just n. That was from our last lecture, the kai square distribution with n degrees of freedom. So we know that. We know it's 1, the blue term, the kai square distribution, n minus 1 degrees of freedom. And we know that it's 2, the green term, follows a kai square distribution with 1 degree of freedom. and we also know x3 is equal to x1. If we have kye squared random variable, another kai squared random variable, sum to a third kai squared random variable with the degrees of freedom added, what does that mean about x1 and x2? It means they're independent. We go back to lecture eight, the second part of this addativity property here. So because of this, we know x1, our blue term, and x2, our green term, are independent. X1, 2 must be independent. x bar is a function two, function, x2, and sigma squared hat are MLE for our variance, the function of x1. It uses the squared deviations around the sample meet. Squared hat, function x1, and from 131, we know A and B are two random variables that are independent. Then any functions of those random variables must also be independent. So then x bar and sigma squared hat are independent as well. Also, questions here. You said that if any two variables are independent and functions of them are independent, but you've shown that the functions are independent, so that it works both ways? So I have shown that this is independent of this, X bar is a function of this, sigma square hat is a function of this. That's what I'm saying. So this is the last property in our theorem for today, independence of the sample mean, and the MLE for the variance. And again, the last thing I'll show is just a simulation to kind of reinforce this idea or convince you if you're not convinced already. Again, we're going to take repeated samples. Looks like a thousand this time. Size n equals 10. My normal population, the true mean is 10. The variance is four. And we're going to compute that sample mean X bar. We're going to compute the sample mean sigma squared a hat each time we take a sample. So we'll have a thousand sample means. We'll have a thousand MLEs for the variance. And we're going to create a scatterploid. of those two things. Each sample is going to be a point, the value for the sample mean on one axis, the value for the MLE of the variance on the other axis, and we'll have a thousand points on this scatter plot. So before I show the scatter plot, what would you expect it to look like if there was dependence, if they were not independent? Yeah, we would expect to see some kind of signal that we could pick up on. Maybe they, as one gets bigger, the other tends to get bigger so they fall on a positive sloping line. Or maybe as one gets bigger, the other tends to get smaller, so they fall on a negative sloping line. Or maybe it's not a linear relationship. but we see some kind of clear relationship. That's what we would look for if they were not independent. And this is what we get. It's just kind of a blob, a cloud of place. There's no clear relationship there. Not like the sample mean gets bigger as the variance gets bigger, or vice versa, or one gets smaller as the other gets bigger. And so we see visually here there's no relationship. They appear to be independent. And in fact, we showed mathematically that they are. Any questions? Again, the main

takeaway today is just this first theorem I gave you. Just be very familiar. with these three points. We're going to use them.

We want to know the sampling distribution, the sample mean, and the MLE for the variance when we have a normal population where we don't know either parent. So here we've seen the sample mean is normally distributed with mean mu, the true mean, variance, sigma squared over n. The sample variance, if we scale it appropriately, n over sigma squared, follows a kai squared distribution, n minus one degrees of freedom. We've lost a degree of freedom by plugging in the sample mean, our estimate for me. since we didn't actually know it. And then lastly, these two things are independent. X bar and our MLE of the variance are independent. I'll stop there. We'll talk about the T distribution on Thursday.